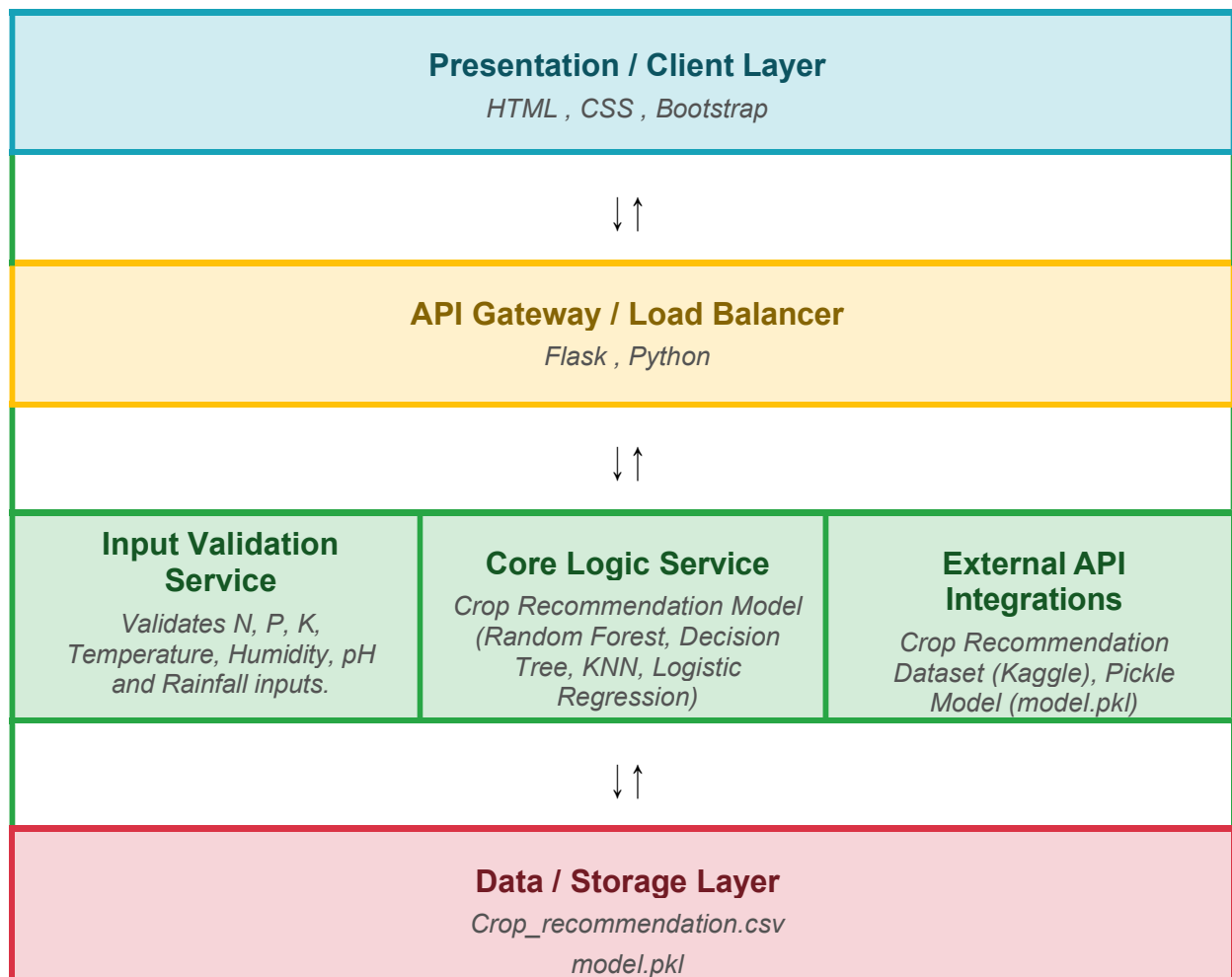


Solution Architecture

Date	26 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for entering soil and weather details and displaying crop recommendations.</i>	<i>HTML, CSS, Bootstrap</i>
<i>Application Layer</i>	<i>Processes user requests and connects the frontend with the machine learning model.</i>	<i>Python, Flask</i>
<i>Machine Learning Component</i>	<i>Predicts the most suitable crop based on environmental conditions.</i>	<i>Scikit-learn, Pandas, NumPy</i>
<i>Data Storage</i>	<i>Stores the agricultural dataset and trained machine learning model.</i>	<i>Crop_recommendation.csv, Pickle (model.pkl)</i>